

Geometry

Unit 7

Lesson 6 Homework

Name _____

Date _____

Period _____

$\triangle YWT$ is a right triangle where $m\angle YTW = 90^\circ$, $m\angle YWT = 30^\circ$, and $YT = 7$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

1.

2.

Side	Length
$\overline{WT}$	
$\overline{YW}$	

$\triangle RXH$ is a right triangle where $m\angle XRH = 90^\circ$, $m\angle RXH = 45^\circ$, and $XR = 12$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

3.

4.

Side	Length
$\overline{RH}$	
$\overline{XH}$	

$\triangle KGM$ is a right triangle where $m\angle MGK = 90^\circ$, $m\angle GKM = 60^\circ$, and $KM = 16$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

5.

6.

Side	Length
$\overline{KG}$	
$\overline{GM}$	

$\triangle FND$ is a right triangle where $m\angle FND = 90^\circ$, $m\angle NDF = 45^\circ$, and $DF = 36$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
7.	$\overline{ND}$	
8.	$\overline{NF}$	

$\triangle YBM$ is a right triangle where $m\angle YMB = 90^\circ$, $m\angle MYB = 30^\circ$, and $YB = 24$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
9.	$\overline{MB}$	
10.	$\overline{YM}$	